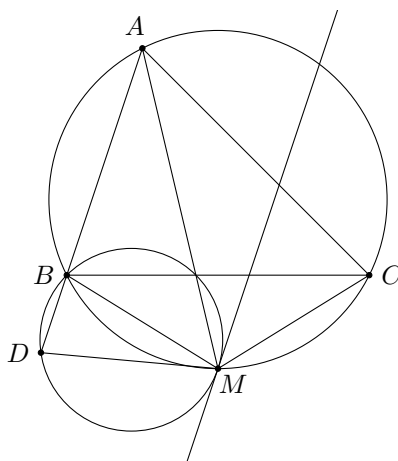




Problem 25 (Nordic MO 2012 Problem 2). In $\triangle ABC$, let M be the midpoint of the arc BC of the circumcircle of $\triangle ABC$ that does not contain A . Let ℓ be the line passing through M and parallel to AB . Suppose $D \neq B$ is a point on the line AB such that the circumcircle of $\triangle BDM$ is tangent to ℓ . Prove that $AC = AD$.



Solution. Since the tangent at M to (BDM) is parallel to BD , we must have $MB = MD$. Thus,

$$\angle MDA = \angle MDB = \angle DBM = \angle MCA,$$

We also have $\angle DAM = \angle CAM$ as M is the midpoint of arc BC of (ABC) . Together with the common side AM , we find that $\triangle ADM \cong \triangle ACM$. Therefore, $AD = AC$. $\square$

Problem 26 (Nordic MO 2022 Problem 1). Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)f(1-x)) = f(x) \quad \text{and} \quad f(f(x)) = 1 - f(x)$$

for all $x \in \mathbb{R}$.

Solution. Answer: $f(x) = \frac{1}{2}$ for all $x \in \mathbb{R}$.

It is easy to check that the constant function $f(x) = \frac{1}{2}$ satisfies both conditions. Now, denote

$$f(f(x)f(1-x)) = f(x), \quad (1)$$

$$f(f(x)) = 1 - f(x). \quad (2)$$

Let S be the range of f . For any $a \in S$, equation (2) gives $f(a) = 1 - a$, and hence $1 - a \in S$. Putting $x = a$ in (1) and using (2) (with $f(x) = a$ and $f(x) = 1 - a$ respectively), we find that

$$f((1-a)a) = f(f(a)f(1-a)) = f(a) = 1-a \quad (3)$$

for all $a \in S$. Since $1 - a \in S$, we can replace a by $1 - a$. Then (3) becomes

$$f(a(1-a)) = a.$$

Comparing this with (3), we find that $1 - a = a$, and hence $a = \frac{1}{2}$. This shows S cannot contain any value other than $\frac{1}{2}$. In other words, f can only be the constant function $f(x) = \frac{1}{2}$. $\square$

Problem 27 (Kazakhstan MO 2017 Problem 6). Prove that there are infinitely many odd composite positive integers n such that $n \mid 2^{\frac{n-1}{2}} + 1$.

Solution. We claim that $n = \frac{4^p + 1}{5}$ satisfies the conditions where p is a prime number of the form $4k + 1$. There are infinitely many such primes by Dirichlet's theorem.

Note that n is an integer since $4^p + 1 \equiv (-1)^p + 1 = -1 + 1 = 0 \pmod{5}$. Clearly, n is odd. Let $a = 4^k$. Then we have

$$4^p + 1 = 4^{4k+1} + 1 = 4a^4 + 1 = (2a^2 + 2a + 1)(2a^2 - 2a + 1).$$

As $2a^2 + 2a + 1 > 2a^2 - 2a + 1 > a^2 > 5$, we know that $n = \frac{4^p + 1}{5}$ is composite. Finally, as $n \mid 4^p + 1 = 2^{2p} + 1$ and

$$\frac{\frac{n-1}{2}}{2p} = \frac{4^{p-1} - 1}{5p}$$

is an odd integer (by the Fermat little theorem and the fact $(5, p) = 1$), we must have

$$n \mid 2^{2p} + 1 \mid 2^{\frac{n-1}{2}} + 1.$$

This completes the proof. □

Problem 28 (USAMO 2015 Problem 3). Let n be a positive integer, and let $S = \{1, 2, \dots, n\}$. For each of the 2^n subsets of S , we colour the subset black or white. Then for any subset T of S , we denote by $f(T)$ the number of subsets of T which are black. Find the number of ways to colour the subsets such that

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2) \tag{1}$$

for any subsets T_1 and T_2 of S .

Solution. Answer: $3^n + 1$.

Let $\mathcal{F}$ be the family of all subsets T of S such that $f(T) \neq 0$. If $\mathcal{F} = \emptyset$, this means all subsets of S are white. This is clearly one possible colouring. Otherwise, let A be a set in $\mathcal{F}$ with the smallest number of elements. For any other $B \in \mathcal{F}$, if A is not a subset of B , then $A \cap B$ has a smaller size than A . However, from

$$f(A \cup B)f(A \cap B) = f(A)f(B) \neq 0,$$

we see that $A \cap B \in \mathcal{F}$. This is a contradiction. Therefore, A must be a subset of B . On the other hand, for any subset B of S which contains A , it is clear that $f(B) \geq f(A) > 0$ by definition. Thus, the elements in $\mathcal{F}$ are exactly those subsets of S which contain A .

We first show that for every subset C of S , there is exactly one way to colour the subsets of S such that

- (1) holds;
- $\emptyset$ is black (i.e. the above A is $\emptyset$); and
- the subset $\{c\}$ is black iff $c \in C$.

Indeed, once the colouring of the singleton sets are fixed, we can determine at most one value of $f(T)$ for each T by using (1) and induction on $|T|$ (since (1) tells us $f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2)$ for disjoint sets T_1, T_2). This in turn gives us at most one possible colouring. On the other hand, we can colour all subsets of C in black and all other sets in white. It is easy to show that $f(T) = 2^{|T \cap C|}$, and hence

$$f(T_1 \cup T_2)f(T_1 \cap T_2) = 2^{|(T_1 \cup T_2) \cap C| + |(T_1 \cap T_2) \cap C|} = 2^{|T_1 \cap C| + |T_2 \cap C|} = f(T_1)f(T_2).$$

This proves our claim.

Now, we have shown that there are 2^n colourings with $A = \emptyset$. In general, for an arbitrary A with $|A| = k$, since $B \in \mathcal{F}$ iff B contains A , the only possible black subsets are those containing A . Thus, we may ignore all elements in A from the set S . It is now the same as colouring all subsets of the set containing the remaining $n - k$ elements where $\emptyset$ (obtained from the original set A) is black. From above, there are 2^{n-k} such colourings. Since there are $\binom{n}{k}$ ways to select A with $|A| = k$, the final answer is

$$1 + \sum_{k=0}^n \binom{n}{k} 2^{n-k} = 1 + (1 + 2)^n = 1 + 3^n.$$

□